

CLIMATE ECONOMICS, GREEN FINANCE AND CIRCULAR ECONOMY

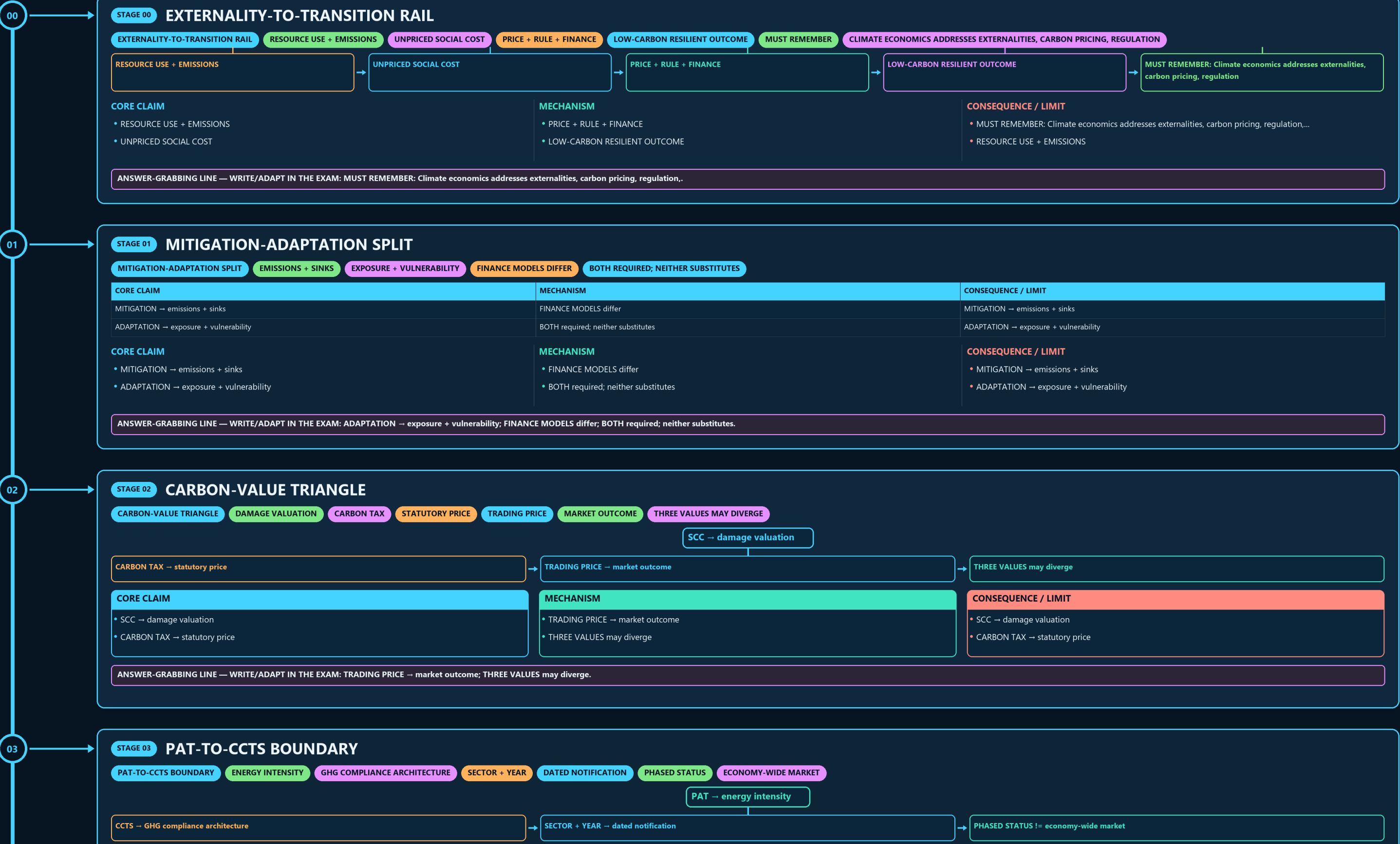
EXTERNALITY-TO-TRANSITION RAIL → MITIGATION-ADAPTATION SPLIT → CARBON-VALUE TRIANGLE → PAT-TO-CCTS BOUNDARY → CLIMATE-RISK CHANNELS → GREEN-FINANCE CREDIBILITY CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• SCC → damage valuation • CARBON TAX → statutory price	• TRADING PRICE → market outcome • THREE VALUES may diverge	• SCC → damage valuation • CARBON TAX → statutory price
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRADING PRICE → market outcome; THREE VALUES may diverge.		

03

STAGE 03 PAT-TO-CCTS BOUNDARY		
PAT-TO-CCTS BOUNDARY	ENERGY INTENSITY	GHG COMPLIANCE ARCHITECTURE
SECTOR + YEAR		
DATED NOTIFICATION		
PHASED STATUS		
ECONOMY-WIDE MARKET		
PAT → energy intensity		
CCTS → GHG compliance architecture	SECTOR + YEAR → dated notification	PHASED STATUS != economy-wide market
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• PAT → energy intensity • CCTS → GHG compliance architecture	• SECTOR + YEAR → dated notification • PHASED STATUS != economy-wide market	• PAT → energy intensity • CCTS → GHG compliance architecture
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR + YEAR → dated notification; PHASED STATUS != economy-wide market.		

04

STAGE 04 CLIMATE-RISK CHANNELS		
CLIMATE-RISK CHANNELS	HAZARD EXPOSURE	POLICY + TECHNOLOGY
CREDIT + MARKET + OPERATIONAL		
HOUSEHOLD + FISCAL TRANSMISSION		
PHYSICAL → hazard exposure		
TRANSITION → policy + technology	CREDIT + MARKET + OPERATIONAL	HOUSEHOLD + FISCAL transmission
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• PHYSICAL → hazard exposure • TRANSITION → policy + technology	• CREDIT + MARKET + OPERATIONAL • HOUSEHOLD + FISCAL transmission	• PHYSICAL → hazard exposure • TRANSITION → policy + technology
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREDIT + MARKET + OPERATIONAL; HOUSEHOLD + FISCAL transmission.		

05

STAGE 05 GREEN-FINANCE CREDIBILITY CHAIN		
GREEN-FINANCE CREDIBILITY CHAIN	OUTPUT + OUTCOME METRIC	ASSURANCE + ADDITIONALITY
ELIGIBILITY / TAXONOMY		
ELIGIBILITY / TAXONOMY	OUTPUT + OUTCOME METRIC	ASSURANCE + ADDITIONALITY
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• ELIGIBILITY / TAXONOMY • OUTPUT + OUTCOME METRIC	• ASSURANCE + ADDITIONALITY • ELIGIBILITY / TAXONOMY	• OUTPUT + OUTCOME METRIC • ASSURANCE + ADDITIONALITY
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALLOCATION; OUTPUT + OUTCOME METRIC; ASSURANCE + ADDITIONALITY.		

06

STAGE 06 BOND-USE BOUNDARY		
BOND-USE BOUNDARY	GREEN BOND	ENVIRONMENTAL USES
SUSTAINABILITY BOND		
GREEN + SOCIAL		
SOVEREIGN ISSUE		
EARMARKED BORROWING		
COMPLETED VERIFIED OUTCOME		
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GREEN BOND → environmental uses	SOVEREIGN ISSUE → earmarked borrowing	GREEN BOND → environmental uses
SUSTAINABILITY BOND → green + social	LABEL != completed verified outcome	SUSTAINABILITY BOND → green + social
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT

- ELIGIBILITY / TAXONOMY
- OUTPUT + OUTCOME METRIC

- ASSURANCE + ADDITIONALITY
- ELIGIBILITY / TAXONOMY

- OUTPUT + OUTCOME METRIC
- ASSURANCE + ADDITIONALITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALLOCATION; OUTPUT + OUTCOME METRIC; ASSURANCE + ADDITIONALITY.

06

STAGE 06

BOND-USE BOUNDARY

BOND-USE BOUNDARYGREEN BONDENVIRONMENTAL USESSUSTAINABILITY BONDGREEN + SOCIALSOVEREIGN ISSUEEARMARKED BORROWINGCOMPLETED VERIFIED OUTCOME

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GREEN BOND → environmental uses	SOVEREIGN ISSUE → earmarked borrowing	GREEN BOND → environmental uses
SUSTAINABILITY BOND → green + social	LABEL != completed verified outcome	SUSTAINABILITY BOND → green + social

CORE CLAIM

- GREEN BOND → environmental uses
- SUSTAINABILITY BOND → green + social

MECHANISM

- SOVEREIGN ISSUE → earmarked borrowing
- LABEL != completed verified outcome

CONSEQUENCE / LIMIT

- GREEN BOND → environmental uses
- SUSTAINABILITY BOND → green + social

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOVEREIGN ISSUE → earmarked borrowing; LABEL != completed verified outcome.

07

STAGE 07

CIRCULAR-VALUE LADDER

CIRCULAR-VALUE LADDERREDUCE + REDESIGNREUSE + REPAIRREFURBISH + REMANUFACTURERECYCLE RESIDUAL MATERIALS

REDUCE + REDESIGN

REUSE + REPAIRREFURBISH + REMANUFACTURERECYCLE residual materials

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• REDUCE + REDESIGN	• REFURBISH + REMANUFACTURE	• REDUCE + REDESIGN
• REUSE + REPAIR	• RECYCLE residual materials	• REUSE + REPAIR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REUSE + REPAIR; REFURBISH + REMANUFACTURE; RECYCLE residual materials.

08

STAGE 08

EPR RESULTS CHAIN

EPR RESULTS CHAINPRODUCER OBLIGATIONCERTIFICATE + VERIFICATIONACTUAL MATERIAL OUTCOME

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• PRODUCER OBLIGATION	• CERTIFICATE + VERIFICATION	• PRODUCER OBLIGATION
• COLLECTION / PROCESSING	• ACTUAL MATERIAL OUTCOME	• COLLECTION / PROCESSING

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COLLECTION / PROCESSING; CERTIFICATE + VERIFICATION; ACTUAL MATERIAL OUTCOME.

09

STAGE 09

JUST-TRANSITION MAP

JUST-TRANSITION MAPWORKERS + SKILLSREGIONS + REVENUECONSUMERS + AFFORDABILITYENERGY ACCESS + SOCIAL PROTECTIONCLOSE DISTINCTIONCARBON TAX IS NOT EMISSIONS TRADING, GREEN LABEL IS

WORKERS + SKILLSREGIONS + REVENUECONSUMERS + AFFORDABILITYENERGY ACCESS + SOCIAL PROTECTIONCLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• WORKERS + SKILLS	• CONSUMERS + AFFORDABILITY	• CLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified...
• REGIONS + REVENUE	• ENERGY ACCESS + SOCIAL PROTECTION	• WORKERS + SKILLS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified.

09

STAGE 09 JUST-TRANSITION MAP

JUST-TRANSITION MAP

WORKERS + SKILLS

REGIONS + REVENUE

CONSUMERS + AFFORDABILITY

ENERGY ACCESS + SOCIAL PROTECTION

CLOSE DISTINCTION

CARBON TAX IS NOT EMISSIONS TRADING, GREEN LABEL IS

WORKERS + SKILLS

REGIONS + REVENUE

CONSUMERS + AFFORDABILITY

ENERGY ACCESS + SOCIAL PROTECTION

CLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified

CORE CLAIM

- WORKERS + SKILLS
- REGIONS + REVENUE

MECHANISM

- CONSUMERS + AFFORDABILITY
- ENERGY ACCESS + SOCIAL PROTECTION

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified...
- WORKERS + SKILLS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Carbon tax is not emissions trading, green label is not verified.

10

STAGE 10 TRADE-CLIMATE BOUNDARY

TRADE-CLIMATE BOUNDARY

EU CBAM

EXTERNAL IMPORT RULE

INDIA CCTS

DOMESTIC SYSTEM

UNSETTLED UNLESS VERIFIED

TRADE PRESSURE

POLICY IDENTITY

EU CBAM → external import rule

INDIA CCTS / MRV → domestic system

RECOGNITION → unsettled unless verified

TRADE PRESSURE != policy identity

CORE CLAIM

- EU CBAM → external import rule
- INDIA CCTS / MRV → domestic system

MECHANISM

- RECOGNITION → unsettled unless verified
- TRADE PRESSURE != policy identity

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State taxonomy/method, boundary, baseline,...
- EU CBAM → external import rule

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State taxonomy/method, boundary, baseline,.

11

STAGE 11 CLIMATE-ECONOMICS ANSWER SPINE

CLIMATE-ECONOMICS ANSWER SPINE

DEFINE METRIC + STATUS

TRACE PRICE + FINANCE + INNOVATION

TEST DISTRIBUTION + ADDITIONALITY

CONCLUDE CREDIBLE JUST TRANSITION

DEFINE metric + status

TRACE price + finance + innovation

TEST distribution + additionality

CONCLUDE credible just transition

CORE CLAIM

- DEFINE metric + status
- TRACE price + finance + innovation

MECHANISM

- TEST distribution + additionality
- CONCLUDE credible just transition

CONSEQUENCE / LIMIT

- DEFINE metric + status
- TRACE price + finance + innovation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TEST distribution + additionality; CONCLUDE credible just transition.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

THE SOCIAL COST OF CARBON VALUES MARGINAL CLIMATE DAMAGE

GREEN TAXONOMIES REDUCE INFORMATION ASYMMETRY WHILE RIGID LABELS MAY

CLIMATE STRESS TESTING EXPLORES SCENARIO EXPOSURE RATHER THAN PREDICTING

BORDER CARBON MEASURES CONNECT DOMESTIC CLIMATE AMBITION WITH TRADE

CIRCULAR BUSINESS MODELS INCLUDE PRODUCT-AS-SERVICE, INDUSTRIAL SYMBIOSIS, REPAIR

ADAPTATION FINANCE HAS PUBLIC-GOOD CHARACTERISTICS AND WEAKER DIRECT REVENUE

MINISTRY OF ENVIRONMENT, FOREST AND CLIMATE CHANGE

OPTIONAL SOURCE DEPTH

- CAUTION: The social cost of carbon values marginal climate damage but depends on discounting and
- CAUTION: Green taxonomies reduce information asymmetry while rigid labels may exclude credible
- CAUTION: Climate stress testing explores scenario exposure rather than predicting one exact
- CAUTION: Border carbon measures connect domestic climate ambition with trade competitiveness and

ADVANCED DEBATE

- CAUTION: Circular business models include product-as-service, industrial symbiosis, repair
- CAUTION: Adaptation finance has public-good characteristics and weaker direct revenue streams
- Ministry of Environment, Forest and Climate Change: coordinates climate policy and
- Ministry of New and Renewable Energy and power-sector institutions: support clean

LEGACY / NUANCE

- RBI and SEBI: shape climate-risk disclosure, green debt and financial-sector
- Urban local bodies and pollution-control systems: implement waste, recycling and
- CAUTION: A carbon price can reduce emissions efficiently but raise household energy costs unless
- CAUTION: A green bond finances defined environmental uses; a sustainability bond can combine

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.